

SUMMER INTERNSHIP REPORT

Artificial Intelligence Internship

Movie Recommendation System & AI Sentiment Analysis Dashboard

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Codec Technologies Pvt. Ltd. — National Internship Portal | Batch 24-28

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Internship Overview

1 Month | AICTE & ICAC Approved

ORGANISATION

Codec Technologies Pvt. Ltd.

PLATFORM

National Internship Portal (Ministry of Education)

PARTNER

Google for Education

ROLE

Artificial Intelligence Intern

DURATION

22 June 2026 – 22 July 2026

The internship followed a project-based learning model: two independent AI/ML applications were designed, built, and deployed end-to-end using Python and Streamlit.

PROJECT 1

Movie Recommendation System

A collaborative-filtering based recommendation engine built on the MovieLens dataset, deployed as a live interactive web app.

100,836

RATINGS

9,742

MOVIES

610

USERS

0.9754

RMSE

How it works

- 1 Merge ratings and movies data
- 2 Build a 610 x 9,719 user-item matrix
- 3 Compute cosine similarity between users
- 4 Recommend top unseen movies by weighted rating

Tech stack

Python

Streamlit

Pandas / NumPy

Scikit-learn

AI Sentiment Analysis Dashboard

A dual-model sentiment analysis tool supporting real-time text and bulk CSV processing with live visualization.

1

Dual-model support

Switch between NLTK VADER (rule-based) and Hugging Face DistilBERT (transformer)

2

Real-time analysis

Instant sentiment scoring for single sentences or paragraphs

3

Bulk CSV uploads

Process thousands of reviews with Plotly pie charts and histograms

4

Data export

Download the fully labelled, processed dataset

What I took away from this internship

1

Built end-to-end ML applications, from raw data to deployed product

2

Learned collaborative filtering and similarity-based recommendations

3

Applied both classical and transformer-based NLP for sentiment analysis

4

Gained hands-on experience with Streamlit for rapid app development

5

Improved data visualization skills using Plotly

6

Practised the full deployment workflow with Git, GitHub and Streamlit Cloud

Thank you

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GitHub — github.com/naitik2424